Pranav Karawale

Precision *beyond* pixels.

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SUMMARY

Founding Engineer and CS student building edge-deployed IoT pipelines and applied ML systems for industrial clients. Proven track record of scaling consumer applications and optimizing cloud infrastructure.

SKILLS

Languages: Python, TypeScript, Go, C, Bash
Backend: Node.js, FastAPI, PostgreSQL, MongoDB, MQTT
Infra & DevOps: AWS (EC2, S3), Docker, Traefik, Grafana
AI & Edge: YOLO, OpenCV, Raspberry Pi, STM32, Modbus/PLC

EXPERIENCE

Pravideon Pvt Ltd — **Jan. 2024 – Present**
Founding Engineer — Navi Mumbai, India

- Deployed custom YOLO vision model (0.77 mAP@50) on edge hardware, integrated with PLCs via Modbus for real-time fire monitor pump control.
- Built Node.js and PostgreSQL metadata aggregator indexing 15M+ records in under 24 hours.
- Developed real-time IoT monitoring dashboard deployed across 10+ industrial sites, achieving 99.9% uptime.
- Created AWS-hosted ingestion pipeline processing and indexing 200K+ rich-media questions for enterprise LMS.

Avignon Université — **Jun. 2025 – Aug. 2025**
Research Intern — Avignon, France

- Developed LoRaPWN, a MAC-layer security enhancement using ChaCha20-Poly1305 cryptography for LoRaWAN.
- Designed STM32 hardware testbed, achieving 10x cryptographic throughput increase and 90% energy reduction vs standard AES.
- Co-authored research on architecture and energy benchmarks for IEEE submission.

Freelancing — **Apr. 2023 – Dec. 2023**
AI Tooling Engineer — Remote

- Built custom LLM orchestration framework using React JSX as a declarative prompt-structuring syntax.
- Architected cross-model abstraction layer supporting multi-provider tool calling (OpenAI, Anthropic, Gemini).

EDUCATION

Vishwaniketan's iMEET (University of Mumbai) — **Nov. 2022 – Jun. 2026**
B.E. Computer Engineering

PROJECTS

Bratify https://bratify.vercel.app
Interactive Svelte image generator scaled to 30K MAU; achieved perfect Lighthouse scores via optimized rendering.

Aetherline
AI digitization engine for P&ID diagrams utilizing Python, OpenCV, and YOLO.

AccessEye
Automated ALPR and entry-management system with <700ms inference on Raspberry Pi Zero 2 W.

ACHIEVEMENTS

Winner: Echelon '25 Hackathon (AI P&ID Digitization)
Gold Medallist: JuniorSkills India 2021 (Web Technologies - National Level)
Leadership: Technical Lead, GDG on Campus VIMEET (2024-25)
OSS Contributor: Astro [https://astro.build]